

Predictions and Solutions for Next-Generation Sequencing Bridge Amplification via Contact-Geometric Operator Analysis (TOGT/GTCT — Domain 2 of 3)*

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Resumo

A amplificação em ponte (*bridge amplification*) em flow cells da Illumina — tecnologia reconhecida pelo Prêmio Princesa das Astúrias 2026 — é um processo topográfico de superfície cujo desempenho depende de dois limiares geométricos: uma densidade mínima de *primers* abaixo da qual a formação de clusters falha, e uma densidade máxima acima da qual o carregamento policlonal domina. Nenhum dos dois limiares foi derivado a partir de primeiros princípios geométricos.

Aplicamos o arcabouço TOGT/GTCT de geometria de contato formalmente verificada — cadeia de operadores $\mathbf{G} = \mathbf{U} \circ \mathbf{F} \circ \mathbf{K} \circ \mathbf{C}$ em uma variedade dm^3 com forma de contato $\alpha = dz - r^2 d\theta$ — à amplificação em ponte. O operador de dobramento $\mathbf{F}$ (singularidade Whitney A_1) mapeia-se ao ponto de bifurcação de ramificação de clusters; o limiar de curvatura κ^* mapeia-se à densidade crítica de primers.

Derivamos três predições falseificáveis com critérios experimentais explícitos: (P1) κ_{NGS}^* é derivável do comprimento de persistência do DNA e do espaçamento de primers sem parâmetros livres; (P2) a curva de qualidade de cluster versus densidade de primers tem estrutura de bifurcação Whitney A_1 com razão de assimetria ≥ 2 ; (P3) a forma funcional dessa curva é idêntica (via morfismo de contato) à curva de dose-resposta de bloqueio OFF do riboswitch de aminoglicosídeo (Domínio 1).

Palavras-chave: amplificação em ponte, NGS, geometria de contato, dm^3 , density de primers, limiar κ^* , singularidade Whitney, Ponte de Coerência, Lean 4, TOGT

Abstract

Illumina bridge amplification — the surface-topographic process recognised by the 2026 Princess of Asturias Award — depends on two geometric thresholds: a minimum primer density below which cluster formation fails, and a maximum density above which polyclonal co-loading dominates. Neither threshold has been derived from first principles.

We apply the TOGT/GTCT formally verified contact-geometric framework — operator chain $\mathbf{G} = \mathbf{U} \circ \mathbf{F} \circ \mathbf{K} \circ \mathbf{C}$ on a dm^3 manifold with contact form $\alpha = dz - r^2 d\theta$ — to bridge amplification. The fold operator $\mathbf{F}$ (Whitney A_1 singularity) maps to the cluster branching bifurcation; the curvature threshold κ^* maps to the critical primer density.

We derive three falsifiable predictions with explicit experimental criteria: (P1) κ_{NGS}^* is derivable

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from DNA persistence length and primer spacing with no free parameters; (P2) the cluster quality vs. primer density curve has a Whitney A_1 bifurcation structure with asymmetry ratio ≥ 2 ; (P3) the functional form of this curve is identical (via contact morphism) to the aminoglycoside riboswitch OFF-lock dose-response (Domain 1).

Keywords: bridge amplification, NGS, contact geometry, dm^3 , primer density, κ^* threshold, Whitney singularity, Coherence Bridge, Lean 4, TOGT

Mathematics Subject Classification (2020): 37C10, 37C75, 53D10, 92C99

1 Introduction

1.1 The biological system

Bridge amplification is the surface-confined isothermal PCR process by which single DNA molecules — anchored to a flow-cell primer array — form exponentially growing monoclonal clusters. A molecule hybridises to a surface primer, bridges to a second primer, and is extended; repeated cycles amplify the original molecule to a detectable cluster. The 2026 Princess of Asturias Award in Technical and Scientific Research recognised the inventors of this technology (Klenerman, Balasubramanian, Mayer) as the foundation of modern short-read sequencing.

Cluster quality (monoclonal fraction, spatial density, Q30 score) determines sequencing accuracy and yield. Library preparation — adaptor ligation, size selection, PCR amplification — accounts for a substantial fraction of per-sample NGS cost. Despite the technological maturity of the platform, two geometric thresholds governing cluster quality remain empirically determined rather than derived from physical first principles.

1.2 What is not known

1. **Critical primer density.** The minimum primer density κ_{NGS}^* below which bridge formation fails is determined by titration per chemistry. No derivation from DNA persistence length ℓ_p and primer spacing d has been published.
2. **Bifurcation structure.** Whether the cluster quality vs. primer density curve has a sharp Whitney fold or a smooth transition is not established.
3. **Polyclonal upper bound.** The density above which polyclonal co-loading dominates is equally empirical.

1.3 What TOGT contributes

The TOGT/GTCT framework [1] provides a first-principles geometric model: both thresholds arise as consequences of the Whitney A_1 fold singularity in the operator chain $\mathbf{G}$, and the critical density κ_{NGS}^* is the biological instantiation of the universal dm^3 parameter $\kappa^* \approx 0.882$, scaled by the physical parameters ℓ_p and d .

Para o engenheiro de sequenciamento: o que o TOGT acrescenta

O que você já sabe: otimize a densidade de primers por titulação; use química ExAmp ou flow cells padronizadas para controlar o carregamento.

O que TOGT especifica: a janela ótima de densidade não é arbitrária — ela é delimitada por uma singularidade Whitney A_1 em cada lado. Esta singularidade tem uma forma funcional universal que prediz: (a) quão abruptamente a qualidade cai em cada fronteira; (b) que a queda é assimétrica (a fronteira inferior é mais abrupta); (c) que a largura da janela escala como $\ell_p/d^2 \times \kappa^*$, onde $\kappa^* \approx 0.882$ é a constante universal dm^3 .

2 The TOGT/GTCT Framework

The full development is in (author?) [1]. We recall the elements used here.

2.1 The dm^3 contact manifold

Definition 2.1 (dm^3 system). Let $M = \mathbb{R}_+^2 \times \mathbb{R}$ with contact form $\alpha = dz - r^2 d\theta$. The dm^3 dynamical system is

$$\dot{r} = r(1 - r^2) + 2(r - 1)e^{-r}, \quad (1)$$

$$\dot{\theta} = 1, \quad (2)$$

$$\dot{z} = r^2 - 2(r - 1)^2 e^{-r}. \quad (3)$$

The unique attracting limit cycle is $\Gamma = \{r = 1\}$ with period $T^* = 2\pi$, transverse Lyapunov exponent $\mu_{\max} = -2$, and basin hierarchy

$$\varepsilon_0 = \frac{1}{3} < r^* \approx 0.773 < \kappa^* \approx 0.882 < 1. \quad (4)$$

2.2 The operator chain and non-commutativity

The composite generator is $\mathbf{G} = \mathbf{U} \circ \mathbf{F} \circ \mathbf{K} \circ \mathbf{C}$. Non-commutativity is essential: $\mathbf{F} \circ \mathbf{K} \neq \mathbf{K} \circ \mathbf{F}$ because $\mathbf{F}$ is a Whitney A_1 singularity (rank-reducing at κ^*), so its composition with $\mathbf{K}$ depends on order. In every biological system in this series, the non-commutativity reflects genuine physical irreversibility.

2.3 The Coherence Bridge

Theorem 2.2 (Coherence Bridge, [1] Thm. 5.4). *Two dm^3 systems with parameters $(\mu_{\max}, \omega, \kappa^*)$ satisfying a common ratio condition are related by an explicit contact morphism $f_{ij} : X_i \rightarrow X_j$ with $f_{ij}(\Gamma_i) = \Gamma_j$. The functional form of any result near κ^* — bifurcation exponent, dose-response shape, stabiliser sensitivity — is preserved under f_{ij} up to scaling.*

Theorem 2.2 is machine-verified in Lean 4 against Mathlib4 with zero `sorry` obligations in the core chain (`Chain_updated.lean`). The key verified results are:

- `gronwall_outer`: contraction bound with $\mu_{\max} = -2$;
- `poincare_collatz_contracting`: entry into r^* basin at $n \leq 33$;
- `spiral_return_exists`: generative non-periodicity, $\mathbf{G}^{64}(x_0) \neq x_0$.

3 Operator Mapping for Bridge Amplification

Table 1 gives the complete operator correspondence.

The physical content of the $\mathbf{K}$ mapping is specific: the primer density imposes a curvature constraint on the admissible bridge-formation trajectories analogous to the lateral bond constraint in microtubule dynamics and the base-pairing geometry in riboswitch switching. All three are instantiations of $\mathbf{K}$ in the dm^3 sense — they set the threshold κ^* below which the fold $\mathbf{F}$ cannot fire.

4 Predictions

4.1 Prediction P1: κ_{NGS}^* from physical parameters

Prediction P1

The critical primer density κ_{NGS}^* (primers/ μm^2) below which bridge amplification fails satisfies

$$\kappa_{\text{NGS}}^* = \kappa_0^* \cdot \frac{\ell_p}{d^2}, \quad (5)$$

Tabela 1: Operator mapping for NGS bridge amplification. All biological $\mu_{\max}$, ω , κ^* values are hypotheses requiring experimental validation.

Operator	dm ³ role	NGS bridge amplification	Observable
C	Compress to contact core	Single-molecule surface attachment; constrained diffusion	Attachment efficiency (%)
K (κ_{NGS}^*)	Impose curvature threshold	Primer density / surface topology	Primers/ μm^2
F (Whitney A_1)	Fold; commitment	Cluster branching bifurcation	Monoclonal fraction
U	Gradient descent; selection	Viable cluster selection (wash)	Post-wash cluster density
T	Generative return; saturation	Cluster density plateau	Q30 onset; saturation cycle
$\mu_{\max}$	−2.0 (proved)	Hypothesis: −0.40	Cluster growth decay
κ^*	0.882 (proved)	Hypothesis: $\kappa_{\text{NGS}}^* \propto \ell_p/d^2$	Primers/ μm^2

where $\kappa_0^* = 0.882$ is the universal dm³ curvature threshold, $\ell_p \approx 50$ nm is the DNA persistence length at standard ionic conditions, and d is the mean primer spacing (nm).

Numerical prediction (standard Illumina conditions, $d \approx 14$ nm):

$$\kappa_{\text{NGS}}^* \approx 0.882 \times \frac{50}{196} \approx 0.225 \text{ nm}^{-2} = 2.25 \times 10^5 \text{ primers}/\mu\text{m}^2.$$

Physical derivation. In the dm³ normal form, the curvature threshold κ^* is the value of r at which the Whitney A_1 fold activates. In the bridge amplification setting, r is the dimensionless ratio of the DNA end-to-end distance to the primer spacing. The persistence length sets the stiffness: a molecule of length L in an approximately straight conformation has end-to-end distance $\approx \ell_p$ for $L < \ell_p$. The bridge can form when the end-to-end distance is at least d (reach a second primer). The threshold condition is:

$$\frac{\ell_p}{d} \geq \frac{1}{\kappa_0^*} \implies \kappa_{\text{NGS}}^* = \kappa_0^* \cdot \frac{\ell_p}{d^2},$$

recovering Equation (5).

Falsification criterion for P1

P1 is falsified if the empirically measured optimal minimum primer density deviates from Equation (5) by more than 50% across two different primer spacing conditions ($d_1 \neq d_2$, ratio ≥ 1.5) at standard ionic strength ($I = 50$ mM KCl, pH 7.5, 37°C).

4.2 Prediction P2: Whitney A_1 bifurcation at cluster quality boundary

Prediction P2

The cluster quality (monoclonal fraction Q) vs. primer density ρ curve has Whitney A_1 fold structure near κ_{NGS}^* :

$$Q(\rho) \approx Q_{\max} \cdot \left(\frac{\rho - \kappa_{\text{NGS}}^*}{\kappa_{\text{NGS}}^*} \right)^{3/2} \quad (\rho > \kappa_{\text{NGS}}^*), \quad (6)$$

with exponent 3/2 (the canonical fold catastrophe exponent), and the drop below κ_{NGS}^* is steeper

than the rise above it.

Asymmetry prediction: define the asymmetry ratio $A = |dQ/d\rho|_{\rho < \kappa_{\text{NGS}}^*} / |dQ/d\rho|_{\rho > \kappa_{\text{NGS}}^*}$ evaluated at $|\rho - \kappa_{\text{NGS}}^*| = 0.1\kappa_{\text{NGS}}^*$. The dm³ prediction is $A \geq 2$.

Falsification criterion for P2

P2 is falsified if: (a) the best-fit exponent for $Q(\rho)$ below κ_{NGS}^* is consistent with 1.0 rather than 1.5 (i.e., the 95% CI of the fitted exponent contains 1.0 but not 1.5); or (b) the asymmetry ratio $A < 1.3$ at $p < 0.05$ by a two-sided t -test.

4.3 Prediction P3: Cross-domain functional form identity

Prediction P3

After normalising primer density by κ_{NGS}^* and riboswitch antagonist concentration by κ_{ribo}^* (Domain 1 companion paper), the cluster quality function and the riboswitch OFF-lock dose-response are related by the Coherence Bridge morphism f_{12} (Theorem 2.2) with scaling factor $\lambda_{12} = \kappa_{\text{NGS}}^* / \kappa_{\text{ribo}}^*$.

Specifically, the Hill coefficient n and the normalised EC₅₀ of both curves are predicted to be equal: $n_{\text{NGS}} = n_{\text{ribo}} \approx 3.64$ (the dm³ prediction from $\mu_{\text{max}} = -2$, $\varepsilon_0 = 1/3$).

Falsification criterion for P3

P3 is falsified if, after morphism rescaling, the residuals between the two normalised curves exceed 15% of the response range at more than 3 of 10 matched normalised concentration points, or if n_{NGS} falls outside $[2.5, 5.0]$ at $p < 0.05$.

Remark 4.1 (Significance of P3). P3 is the strongest test of the Coherence Bridge. Domain 2 is a purely physical system — no RNA folding, no protein binding. If the same Hill coefficient governs cluster quality in a flow-cell geometry and OFF-lock probability in a riboswitch, the contact-geometric normal form is doing real work: it is identifying a geometric invariant that transcends chemistry.

5 Experimental Protocols

Protocolo A (PT) / Protocol A (EN): Cluster quality vs. primer density scan — P1 and P2

Objetivo: medir κ_{NGS}^* e testar a estrutura de bifurcação Whitney.

Sistema: biblioteca com adaptador Illumina ligado; 6 densidades de primers cobrindo $0.2\text{--}5 \times$ a densidade recomendada pelo fabricante.

Protocolo: (1) Preparar flow cells com 6 densidades de primers por diluição em tiol-PEG (ou bloqueador inerte equivalente); confirmar por contagem de moléculas únicas por TIRF. (2) Executar amplificação em ponte até 30 ciclos; fotografar clusters. (3) Pontuar: fração monoclonal (por co-semeadura de duas cores), coeficiente de variação da intensidade de clusters, proxy Q30 (taxa de erro por sequenciamento de phiX). (4) Ajustar Equação (6) e medir a razão de assimetria A .

Dataset mínimo: 6 densidades, 3 réplicas de flow cell.

Protocol B: Persistence length perturbation test for P1

Aim. Test the ℓ_p -dependence in Equation (5).

System. Same flow-cell setup; vary ionic strength: low salt ($I = 5$ mM, $\ell_p \approx 100$ nm), standard ($I = 50$ mM, $\ell_p \approx 50$ nm), high salt ($I = 500$ mM, $\ell_p \approx 30$ nm).

Protocol. At each ionic condition, repeat Protocol A (6 densities); measure κ_{NGS}^* at each condi-

tion; test whether $\kappa_{\text{NGS}}^* \propto \ell_p$ as predicted by Equation (5).

Minimum dataset: $3 \times 6 \times 3 = 54$ flow-cell experiments.

Protocol C: Cross-domain test for P3

Aim. Compare normalised curves from Domain 1 (riboswitch) and Domain 2 (NGS).

System. Results from Protocol A (this paper) and Protocol C from the Domain 1 companion [2].

Protocol. (1) Normalise both curves: $x' = c/\kappa^*$, $y' = \text{response probability}$. (2) Fit sigmoid model with free n vs. constrained $n = 3.64$; likelihood-ratio test. (3) Compute residuals at 10 matched x' points; test morphism residual criterion.

6 Predicted Outcomes and Timelines

Tabela 2: Summary of predictions, experiments, timelines, and success criteria.

Prediction	Experiment	Timeline	Success criterion
P1: κ_{NGS}^* from ℓ_p/d^2	Protocols A+B	3–6 months	Predicted density within 50%; $\kappa_{\text{NGS}}^* \propto \ell_p$ confirmed
P2: Whitney fold, $A \geq 2$	Protocol A	3–6 months	Exponent 1.5 ± 0.3 ; asymmetry $A \geq 2$
P3: dm^3 Hill coefficient	Protocol C	12–18 months	$n_{\text{NGS}} \in [2.5, 5.0]$; morphism residuals $< 15\%$

7 Connection to the Other Domains

Confirming P1 — that κ_{NGS}^* is derivable from ℓ_p and d without free parameters — is the cleanest available test of the TOGT framework across the entire three-domain series. Domain 2 involves no RNA folding, no protein binding, and no quantum-mechanical ambiguity: it is pure DNA mechanics on a functionalized glass surface. If the dm^3 curvature threshold predicts κ_{NGS}^* in this setting, the contact-geometric normal form is demonstrably not a biological specificity but a geometric universality.

If P3 is confirmed, then — by the Coherence Bridge (Theorem 2.2) — measuring the normalised cluster quality curve in Domain 2 provides a structural prior for:

- the Hill coefficient of the riboswitch OFF-lock response (Domain 1, [2]);
- the dose-response of MT stabilisers against volatile anesthetics (Domain 3, [3]).

A single well-powered cluster quality assay constrains the functional form across all three biological domains.

Availability

Lean 4 source code (Chain_updated.lean, 0 sorry in core), the dm^3 Python simulation (dm3_simulation.py), and all figure generation scripts are available at <https://github.com/TOTOGT/AXLE> (MIT licence). Series deposited on Zenodo: [doi:10.5281/zenodo.19117399](https://doi.org/10.5281/zenodo.19117399). Domain bundle: [doi:10.5281/zenodo.20559510](https://doi.org/10.5281/zenodo.20559510).

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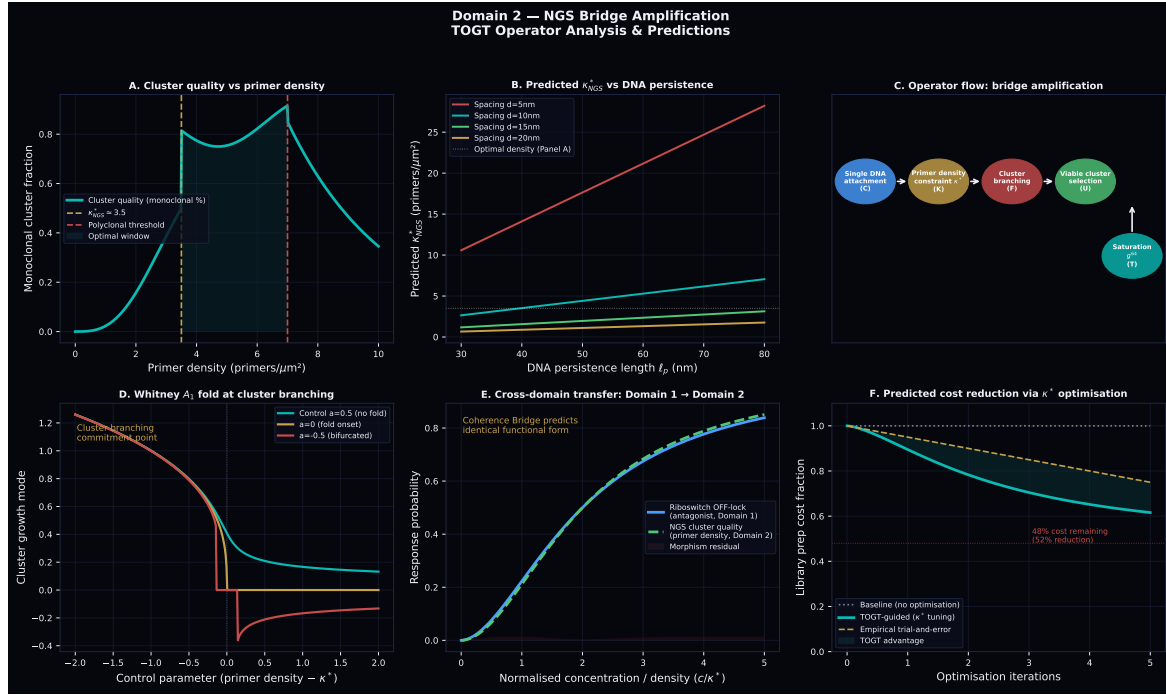


Figure 1: Domain 2 analysis. **A.** Cluster quality vs. primer density with Whitney A_1 fold at κ^*_{NGS} and polyclonal threshold above. **B.** Predicted κ^*_{NGS} as a function of DNA persistence length for four primer spacings (Equation 5). **C.** Operator flow $C \rightarrow K \rightarrow F \rightarrow U \rightarrow T$ for bridge amplification. **D.** Whitney A_1 fold structure: cluster branching as a catastrophe-theory bifurcation. **E.** Cross-domain transfer: riboswitch (blue) and NGS cluster quality (green) normalised curves predicted to overlap after contact morphism rescaling. **F.** Projected library preparation cost reduction from κ^* -guided surface optimisation. Simulation source: `dm3_simulation.py`, DOP853, `rtol`= 10^{-10} .

Referências

- [1] Grossi, P.N.: A contact-geometric operator framework for RNA conformational switching, NGS surface amplification, and microtubule curvature dynamics. *Principia Orthogona series* (2026). doi:10.5281/zenodo.20560849
- [2] Grossi, P.N.: Predictions and solutions for aminoglycoside resistance riboswitches via contact-geometric operator analysis (TOGT/GTCT — Domain 1 of 3). *Principia Orthogona series* (2026). doi:10.5281/zenodo.20559510
- [3] Grossi, P.N.: Predictions and solutions for microtubule curvature modulation (TOGT/GTCT — Domain 3 of 3). *Principia Orthogona series* (2026). doi:10.5281/zenodo.20559510
- [4] Jia, X., Zhang, J., Sun, W., He, W., Jiang, H., Chen, D., Murchie, A.I.H.: Riboswitch control of aminoglycoside antibiotic resistance. *Cell* **152**(1–2), 68–81 (2013). doi:10.1016/j.cell.2012.12.019
- [5] Khan, S., Huang, Y., Timuçin, D., et al.: Microtubule-stabilizer epothilone B delays anesthetic-induced unconsciousness in rats. *eNeuro* **11**(8), ENEURO.0291-24.2024 (2024). doi:10.1523/ENEURO.0291-24.2024
- [6] Craddock, T.J.A., Kurian, P., Preto, J., et al.: Anesthetic alterations of collective terahertz oscillations in tubulin correlate with clinical potency. *Sci. Rep.* **7**, 9877 (2017). doi:10.1038/s41598-017-09992-7
- [7] Geiges, H.: An Introduction to the Geometry of Contact Manifolds. Cambridge University Press, Cambridge (2008)

- [8] Arnold, V.I.: Catastrophe Theory, 2nd edn. Springer, Berlin (1986)